INS 3121
SOIL MECHANICS

Soil Compaction

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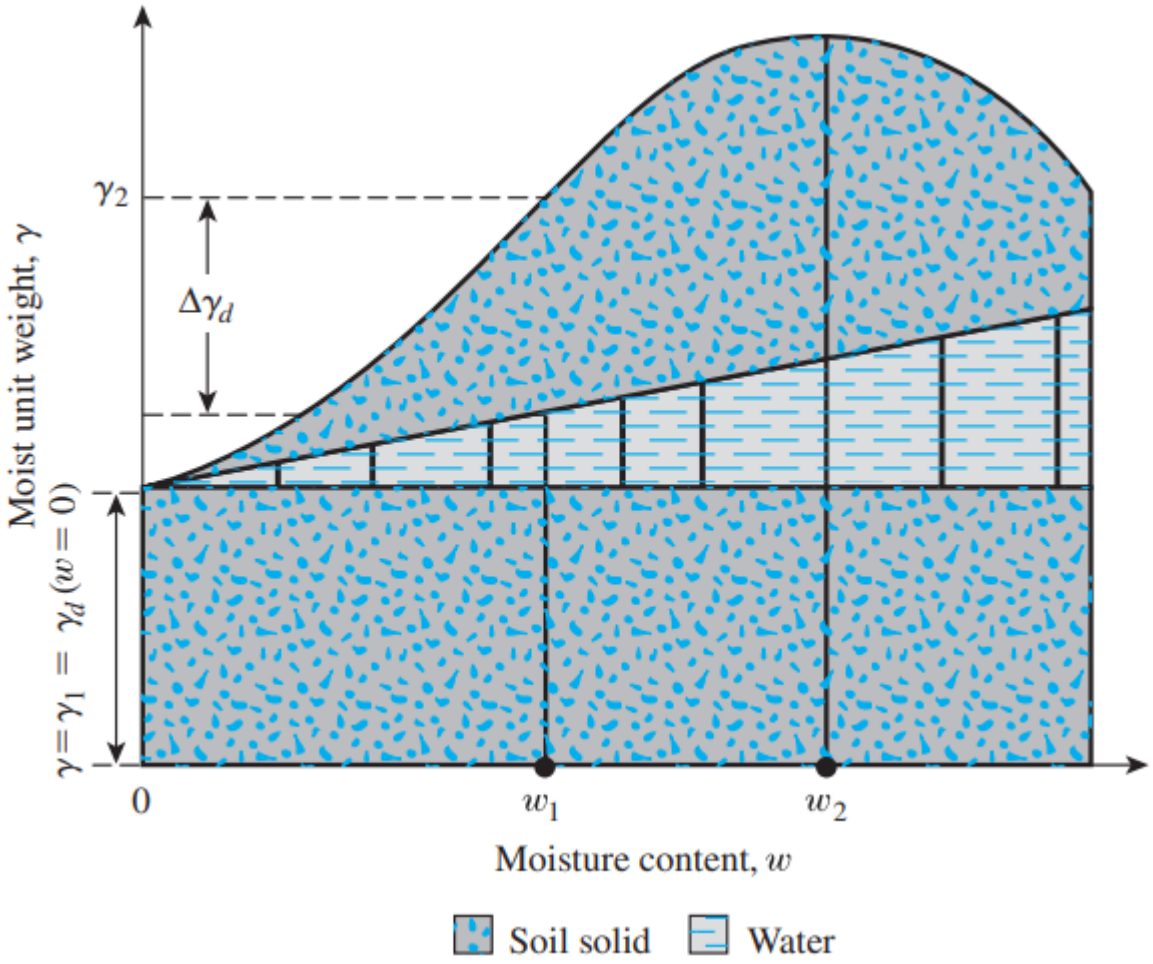
Introduction

- Increase Density
  - increase strength characteristics
  - increase bearing capacity and stability of slopes
  - decrease settlement

  - Roller, vibroflot

Introduction

- Compaction: the densification of soil by removal of air, which requires mechanical energy.
- Degree of compaction is measured in terms of dry unit weight.
- Water
- Optimum Moisture content (OMC): The moisture content at which the maximum dry unit weight is attained

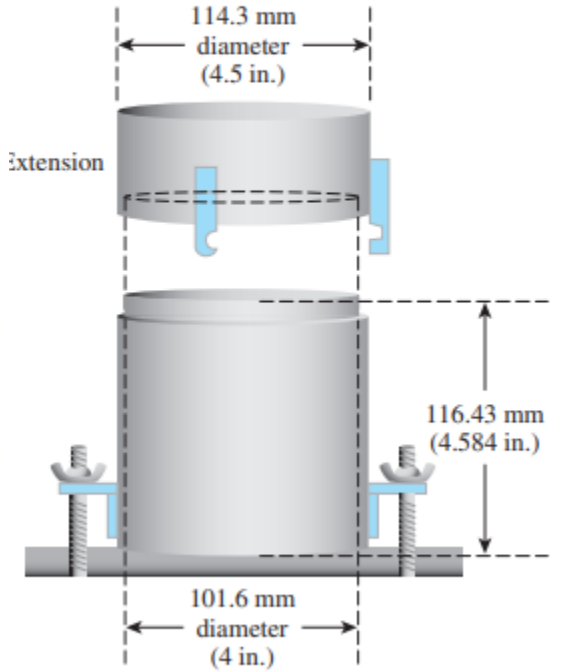


Standard Proctor Test

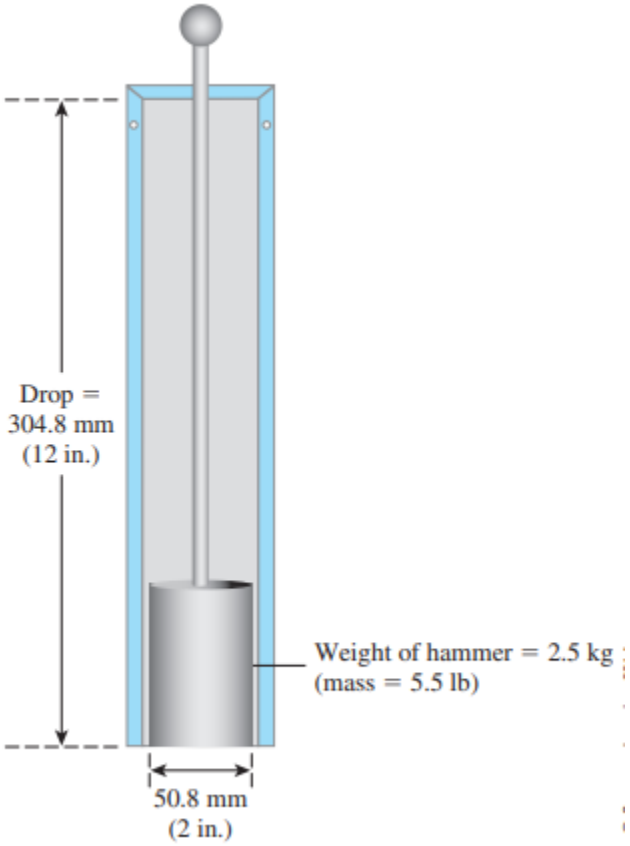
- 25 blows, 2.5 kg hammer, 30 cm drop height
- 3 layers
- Mold V = 944 cm^3, D = 101.6 mm



(c)



(a)



(b)

Standard Proctor Test

$$\gamma = \frac{W}{V_m} \quad \Rightarrow \quad \gamma_d = \frac{\gamma}{1 + \frac{w\,(\%)}{100}}$$

Procedure for the standard Proctor test; elaborated in ASTM Test Designation AASHTO Test Designation T-99

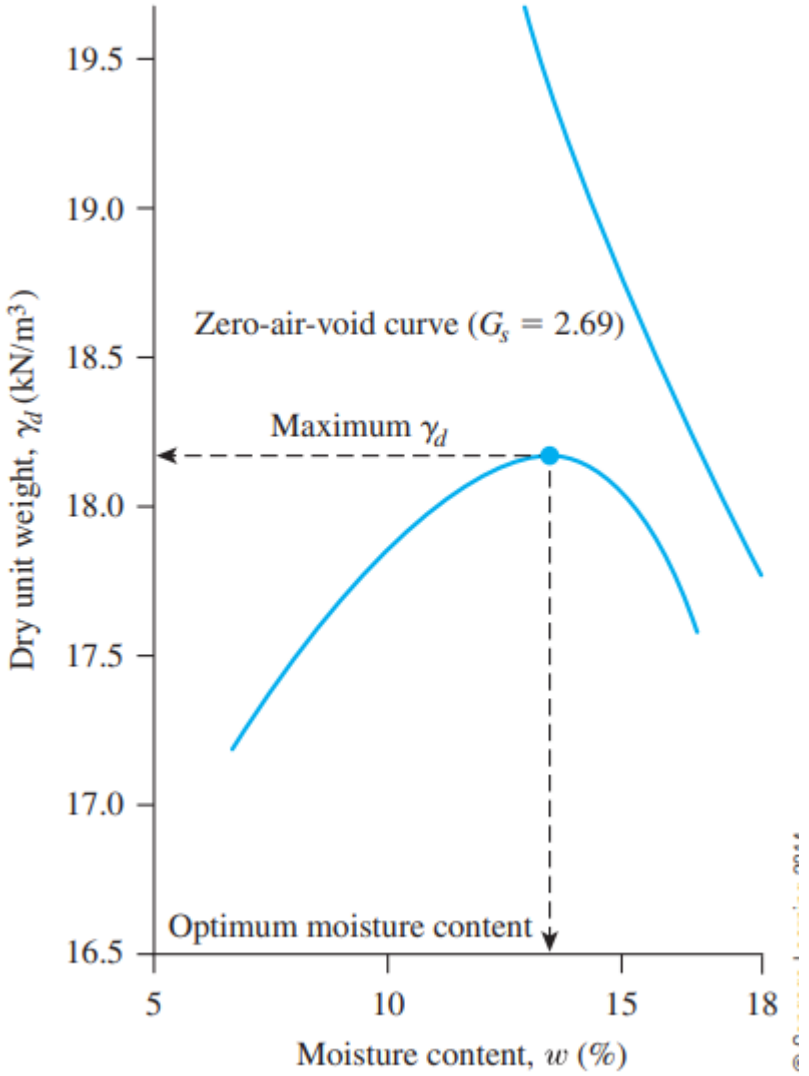
$$\gamma_d = \frac{G_S \gamma_w}{1 + e}$$

For 100% saturation, e = wGs, so

$$\gamma_d = \frac{G_s \gamma_w}{1 + \frac{G_s w}{S}}$$

$$\gamma_{zav} = \frac{G_s \gamma_w}{1 + wG_s} = \frac{\gamma_w}{w + \frac{1}{G_s}}$$

zero air void unit weight



Standard Proctor Test

- To obtain the variation of γ_{zav} with moisture content, use the following procedure:

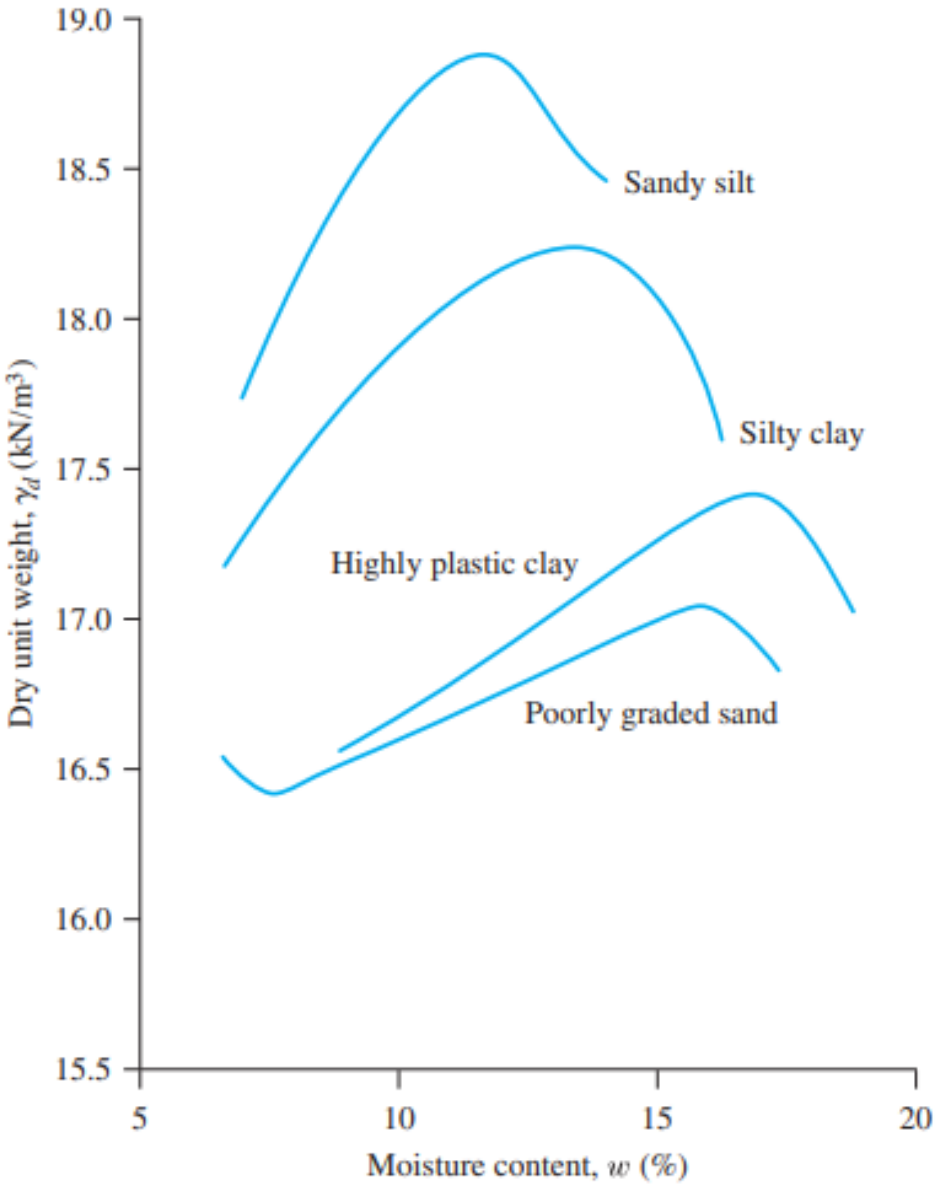
1. Determine the specific gravity of soil solids.
2. Know the unit weight of water (γw).
3. Assume several values of ω, such as 5%, 10%, 15%, and so on.
4. Use the Eq. to calculate γ_{zav} for various values of ω.

Factors Affecting Compaction

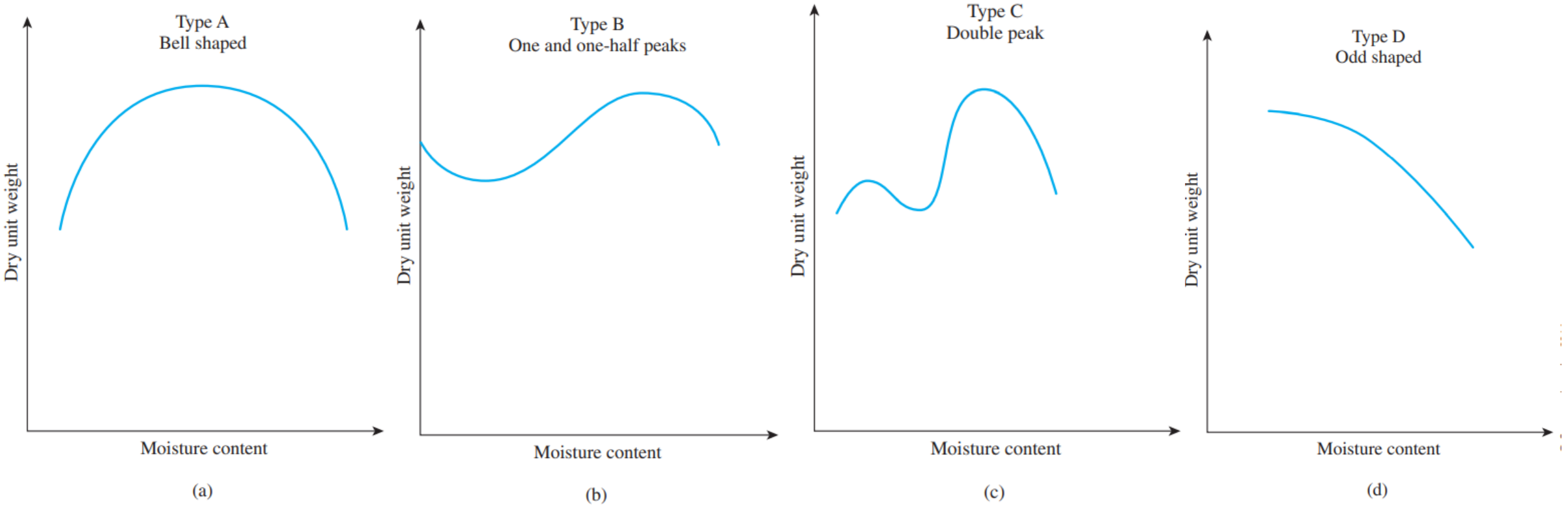
- moisture content

Effect of Soil Type

- grain-size distribution
- shape of the soil grains
- specific gravity of soil solids
- amount and type of clay minerals present



Factors Affecting Compaction



Factors Affecting Compaction

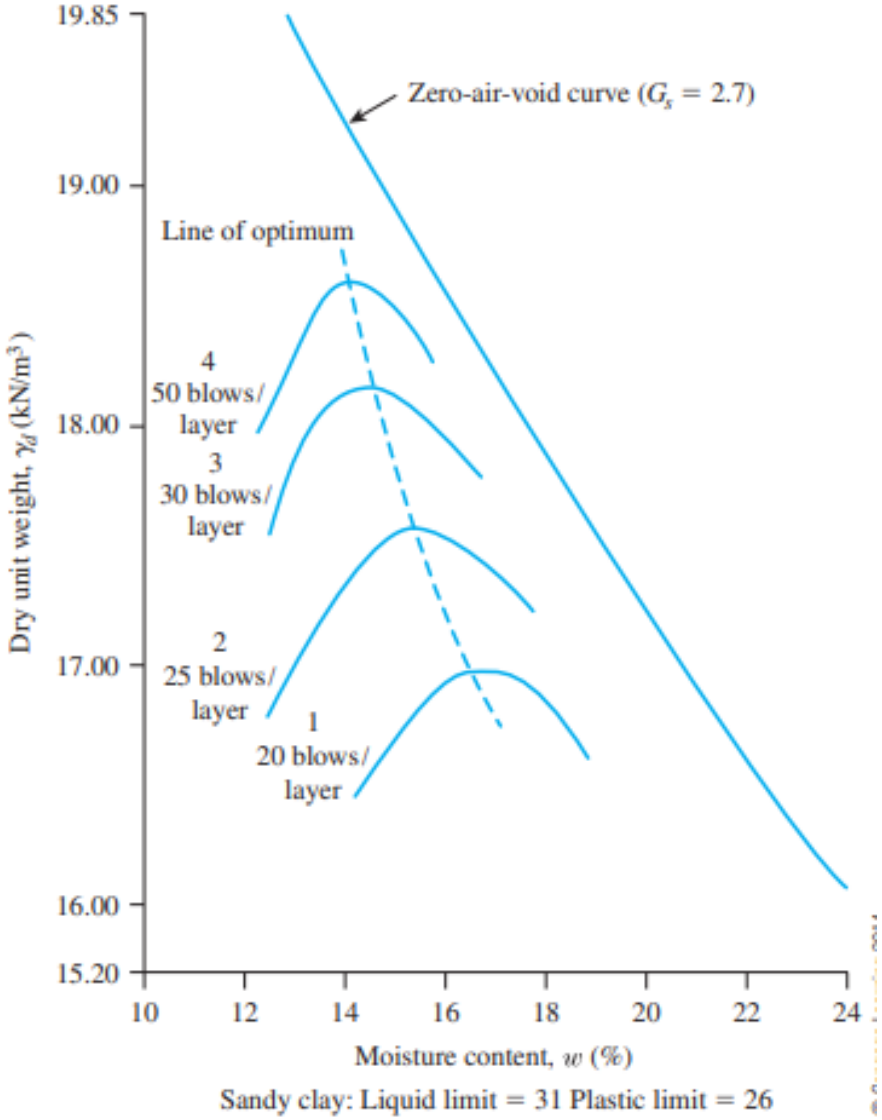
- Effect of Compaction Effort

$$E = \frac{\begin{pmatrix}\text{Number}\\ \text{of blows}\\ \text{per layer}\end{pmatrix} \times \begin{pmatrix}\text{Number}\\ \text{of}\\ \text{layers}\end{pmatrix} \times \begin{pmatrix}\text{Weight}\\ \text{of}\\ \text{hammer}\end{pmatrix} \times \begin{pmatrix}\text{Height of}\\ \text{drop of}\\ \text{hammer}\end{pmatrix}}{\text{Volume of mold}}$$

From the preceding tabulation and Figure, it can be seen that:

1. As the compaction effort is increased, the maximum dry unit weight of compaction is also increased.
2. As the compaction effort is increased, the optimum moisture content is decreased to some extent.

The degree of compaction is not directly proportional to the compaction effort



Modified Proctor Test

- Modified Proctor test (ASTM Test Designation D-1557 and AASHTO Test Designation T-180)
- volume = 944 cm^3
- hammer weight = mass = 4.54 kg
- hammer drop = 457.2 mm
- number of hammer blows = 25
- number of layer = 5

Modified Proctor Test



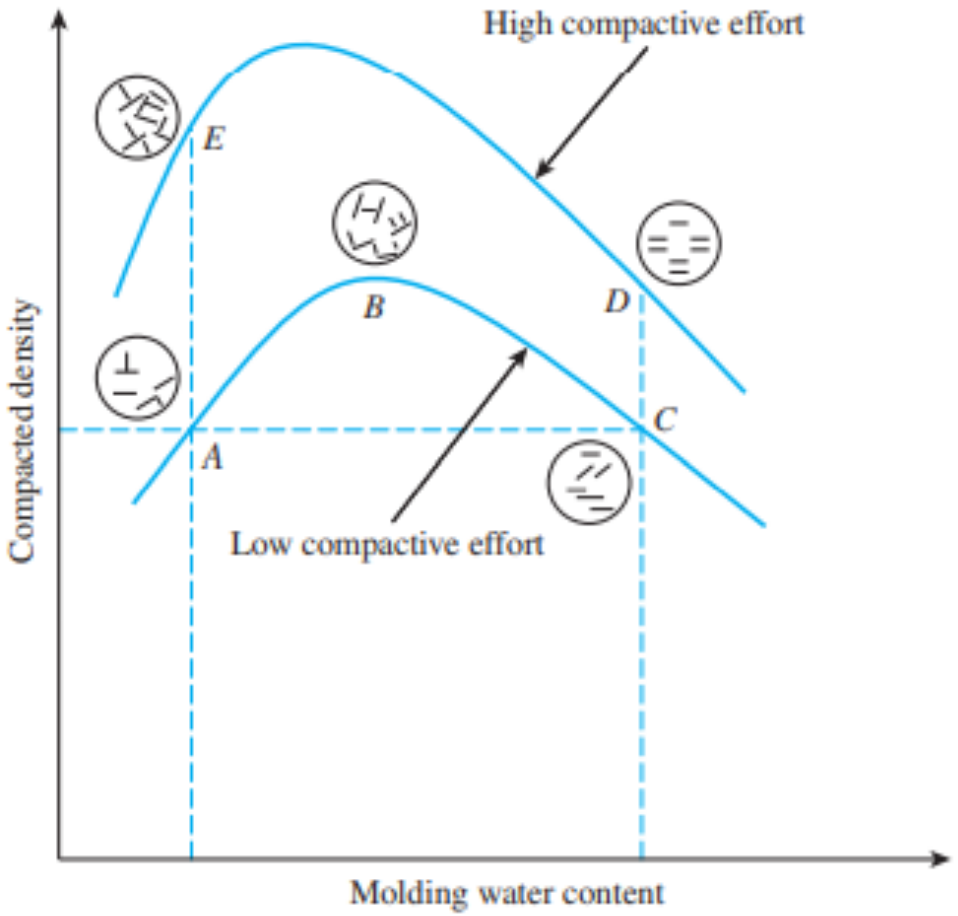
Figure 6.7 Comparison between standard Proctor hammer (left) and modified Proctor hammer (right) (*Courtesy of Braja M. Das, Henderson, Nevada*)

Table 6.1 Summary of Standard and Modified Proctor Compaction Test Specifications (*ASTM D-698 and D-1557*)

	Description	Method A	Method B	Method C
Physical data for the tests	Material	Passing No. 4 sieve	Passing 9.5 mm sieve	Passing 19 mm sieve
	Use	Used if 20% or less by weight of material is retained on No. 4 sieve	Used if more than 20% by weight of material is retained on No. 4 (4.75 mm) sieve and 20% or less by weight of material is retained on 9.5 mm sieve	Used if more than 20% by weight of material is retained on 9.5 mm sieve and less than 30% by weight of material is retained on 19 mm sieve
	Mold volume	944 cm^3	944 cm^3	2124 cm^3
	Mold diameter	101.6 mm	101.6 mm	152.4 mm
	Mold height	116.4 mm	116.4 mm	116.4 mm
Standard Proctor test	Weight of hammer	24.4 N	24.4 N	24.4 N
	Height of drop	305 mm	305 mm	305 mm
	Number of soil layers	3	3	3
	Number of blows/layer	25	25	56
Modified Proctor test	Weight of hammer	44.5 N	44.5 N	44.5 N
	Height of drop	457 mm	457 mm	457 mm
	Number of soil layers	5	5	5
	Number of blows/layer	25	25	56

Structure of Compacted Clay Soil

- If clay is compacted with a moisture content on the dry side of the optimum, it will possess a flocculent structure. (Point A)
- When the moisture content of compaction is increased, lower degree of flocculation and a higher dry unit weight is obtained. (Point B)
- A continued increase in moisture content from B to C expands the double layers more. This expansion results in a more or less dispersed structure. (Point C)



Structure of Compacted Clay Soil

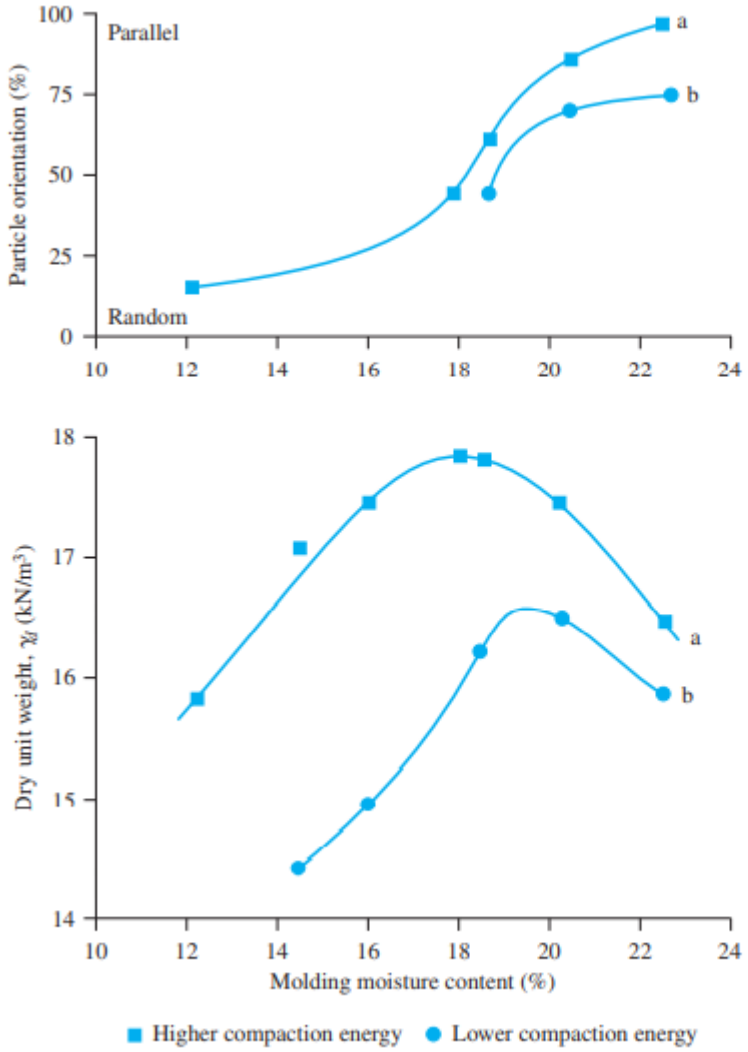
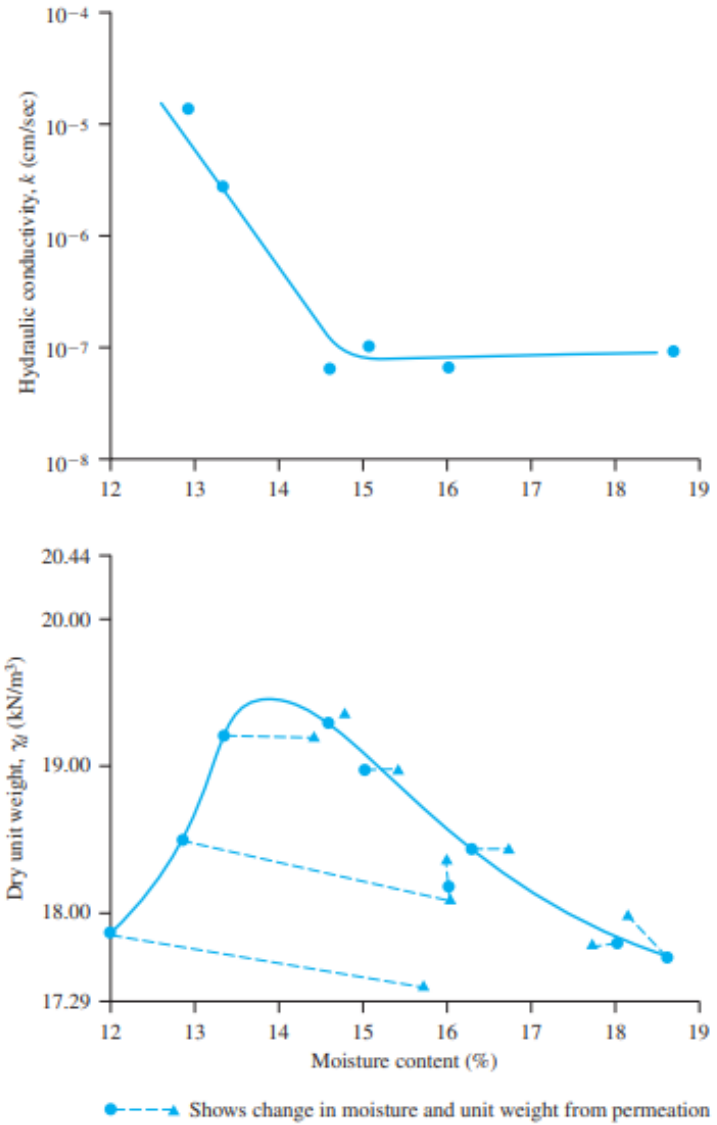


Figure 6.10 Orientation against moisture content for Boston blue clay (*After Lambe, 1958a. With permission from ASCE.*)

Effect of Compaction on Cohesive Soil Properties

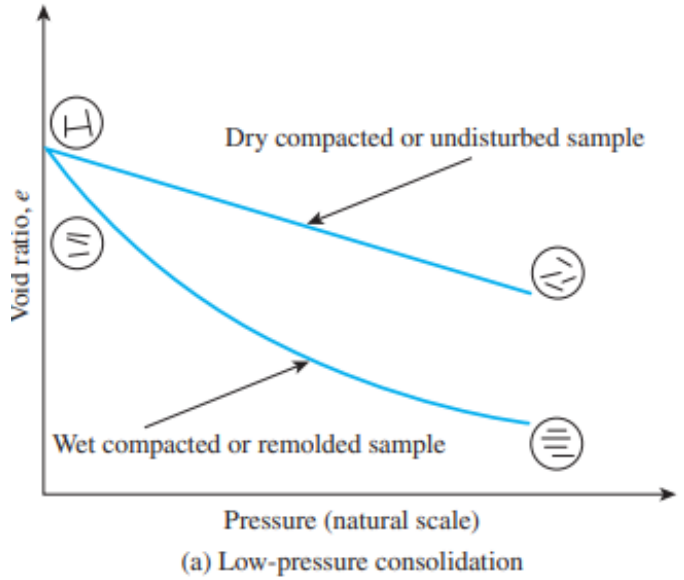
- Compaction induces variation in the structures of cohesive soils.
- Results of these structural variation include changes in hydraulic conductivity, compressibility and strength.
- For instance, according to figure hydraulic conductivity reaches a minimum value at approximately the optimum moisture content.
- Beyond the optimum moisture content, the hydraulic conductivity increases slightly.
- The high value of the hydraulic conductivity on the dry side of the optimum moisture content is due to the random orientation of clay particles that results in larger pore spaces

Effect of Compaction on Cohesive Soil Properties

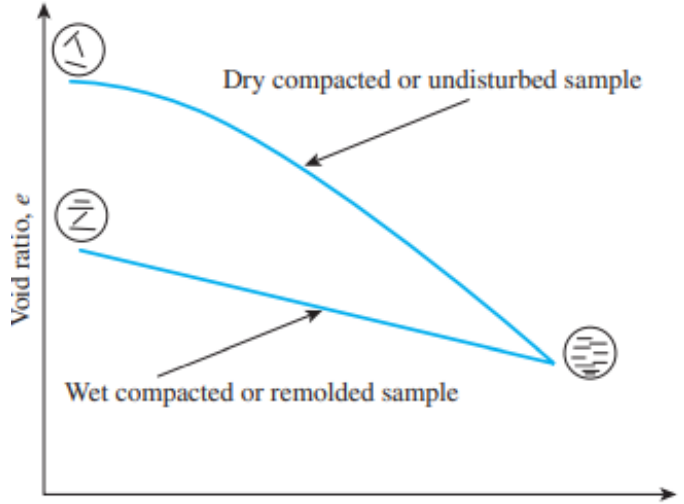


Effect of Compaction on Cohesive Soil Properties

- At low pressure, compressibility of dry side of compacted soil ratio is low and fast. If applied pressure however is large enough to change the soil structure, compressibility ratio of the dry side is higher than wet side.
- At high pressure, when the pressure sufficiently large enough, void ratio of the two samples substantially become the same

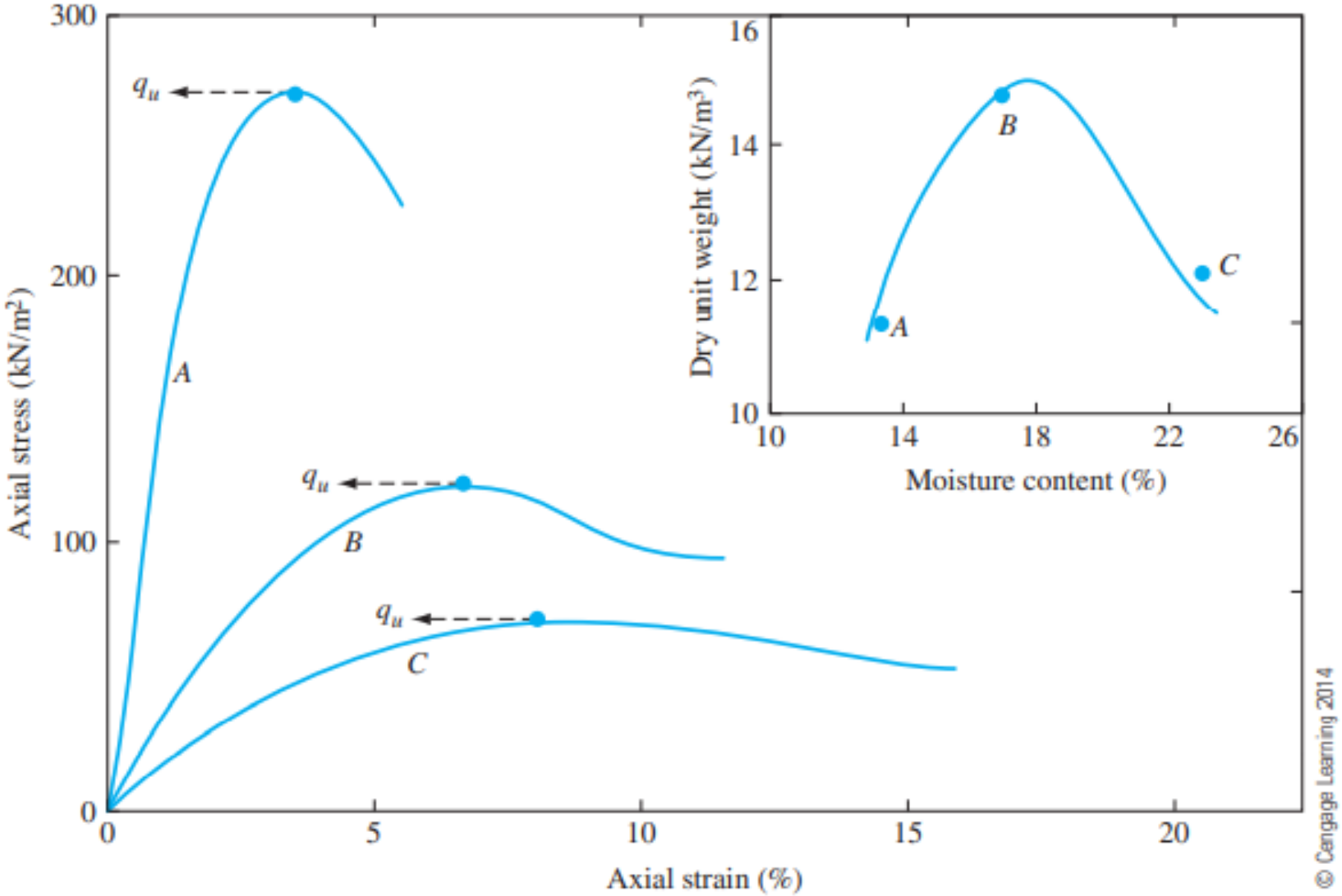


(a) Low-pressure consolidation

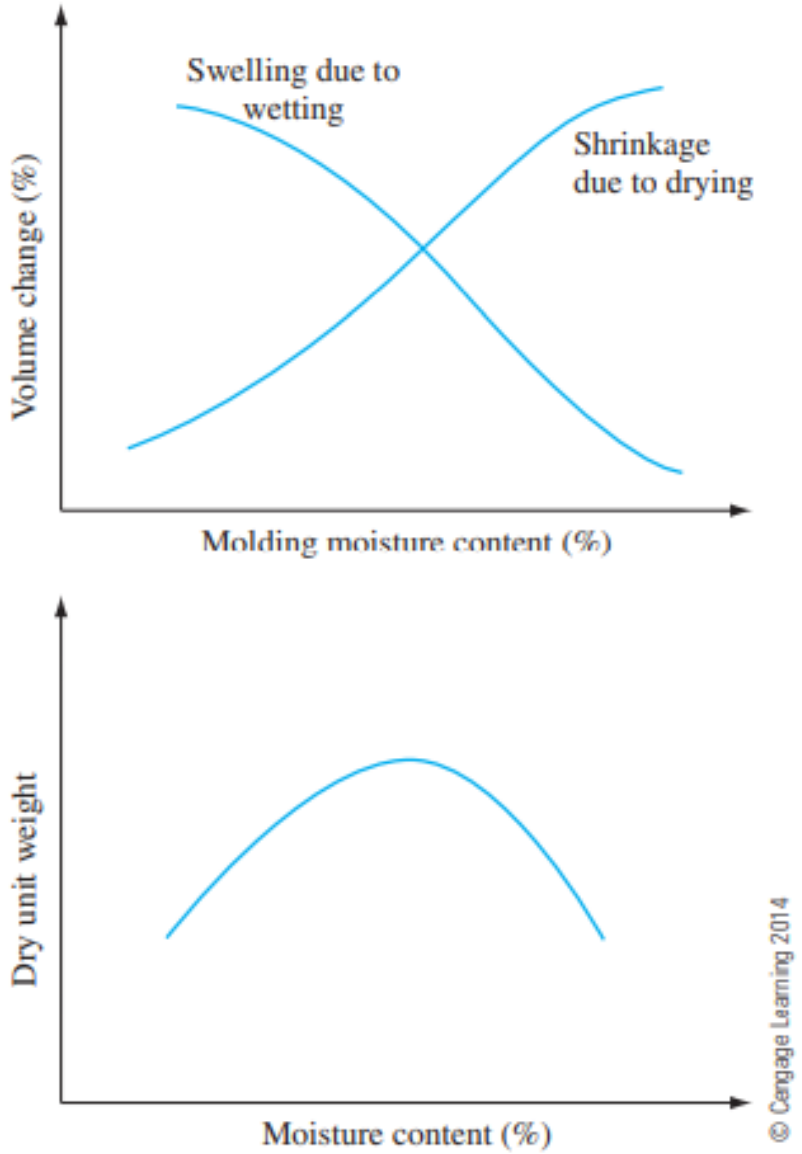


(b) High-pressure consolidation

Effect of Compaction on Cohesive Soil Properties



Effect of Compaction on Cohesive Soil Properties



Field compaction

- the most common types of rollers

1. Smooth-wheel roller(or smooth-drum roller)
2. Pneumatic rubber-tired roller
3. Sheepsfoot roller
4. Vibratory roller

- Smooth-wheel rollers: suitable for proofrolling subgrades and for finishing operation of fills with sandy and clayey soils.
- Contact pressures : 310 – 380 kN/$m2$.

Field compaction



Figure 6.15 Smooth-wheel roller (*Ingram Compaction LLC*)

Field compaction

- Pneumatic rubber-tired rollers: better in many respects than the smooth-wheel rollers.
- Contact pressure : 600 - 700 kN/$m2$
- pneumatic rollers can be used for sandy and clayey soil compaction.
- Compaction is achieved by a combination of pressure and kneading action.



Figure 6.16 Pneumatic rubber-tired roller (*Ingram Compaction LLC*)

Field compaction

- Sheepsfoot rollers : drums with a large number of projections.
- most effective in compacting clayey soils
- contact pressure : 1400 – 7000 kN/$m2$
- During compaction in the field, the initial passes compact the lower portion of a lift.
- Compaction at the top and middle of a lift is done at a later stage.



Figure 6.17 Sheepsfoot roller (*SuperStock/Alamy*)

Field compaction

- Vibratory rollers: very efficient in compacting granular soils
- Vibrators can be attached to smooth-wheel, pneumatic rubber-tired, or sheepsfoot rollers to provide vibratory effects to the soil.
- Hand-held vibrating plates can be used for effective compaction of granular soils over a limited area.



Figure 6.17 Sheepsfoot roller (*SuperStock/Alamy*)

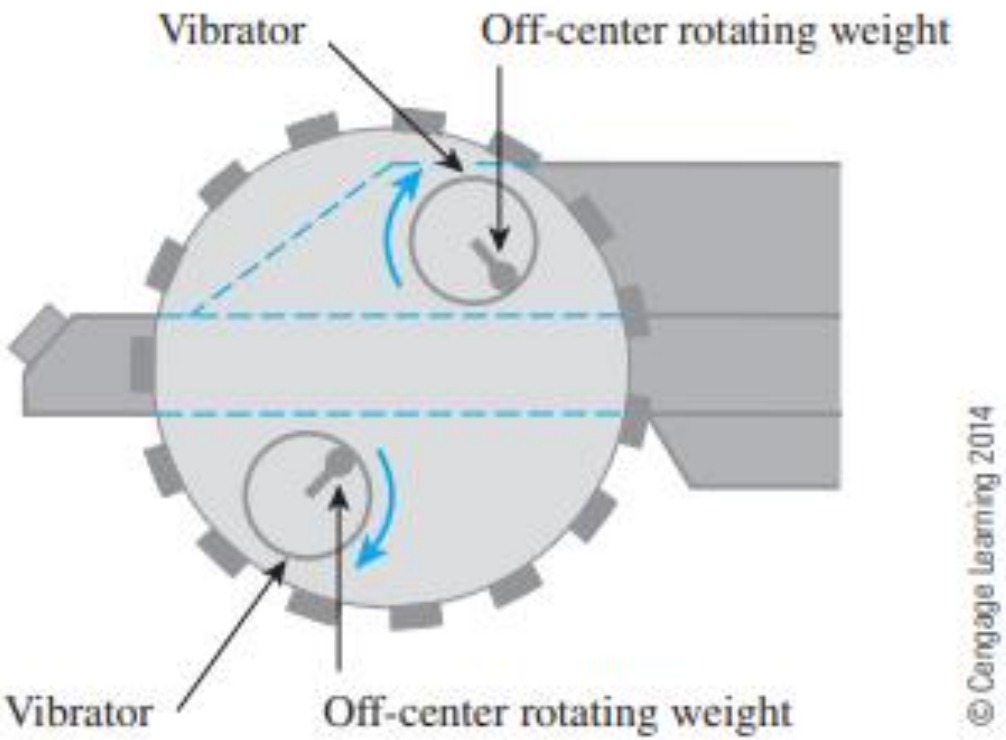
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Figure 6.17 Sheepsfoot roller (*SuperStock/Alamy*)

Field compaction



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Figure 6.18 Principles of vibratory rollers

Field compaction

- Thickness of lift, the intensity and area of pressure applied
- The dry unit weight of a soil at a given moisture content will increase up to a certain point with the number of passes of the roller. Beyond this point it will remain approximately constant. In most cases, about 10 – 15 roller passes yield the maximum dry unit weight economically attainable.

Field compaction

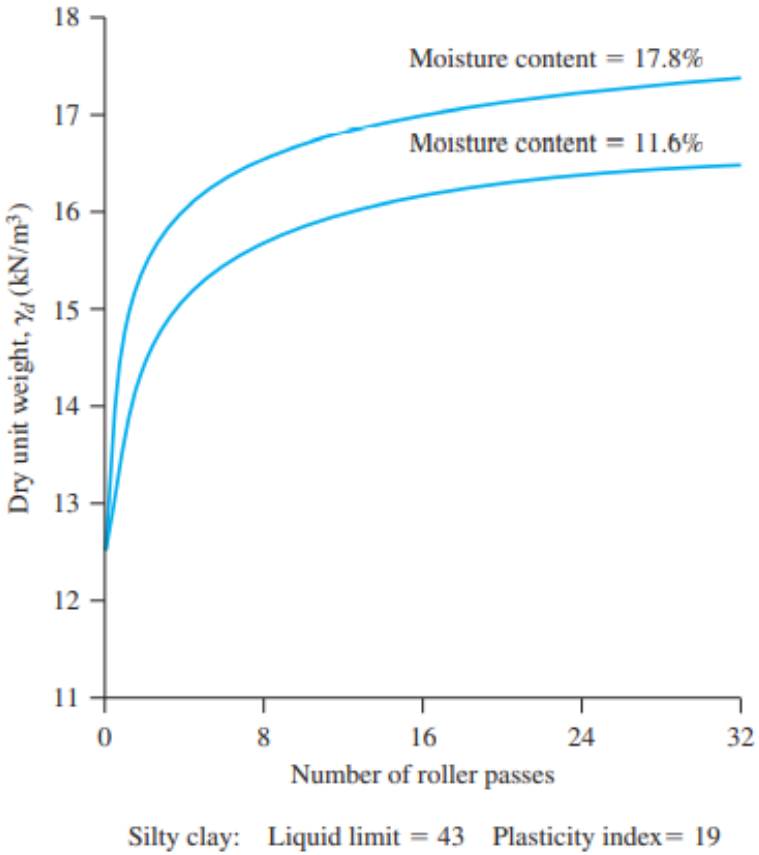


Figure 6.20 Growth curves for a silty clay—relationship between dry unit weight and number of passes of 84.5 kN three-wheel roller when the soil is compacted in 229 mm loose layers at different moisture contents (*From Full-Scale Field Tests on 3-Wheel Power Rollers. In Highway Research Bulletin 272, Highway Research Board, National Research Council, Washington, D.C., 1960, Figure 15, p. 23. Reproduced with permission of the Transportation Research Board.*)

Field compaction

- at any given depth, the dry unit weight of compaction increases with the number of roller passes. However, the rate of increase of unit weight gradually decrease after about 15 passes.
- The dry unit weight and hence the relative density, Dr, reach maximum values at a depth of about 0.5m and gradually decrease at lesser depths
- Once the relationship between depth and relative density (or dry unit weight) for a given soil with a given number of roller passes is determined, it is easy to estimate the approximate thickness of each lift

Field compaction

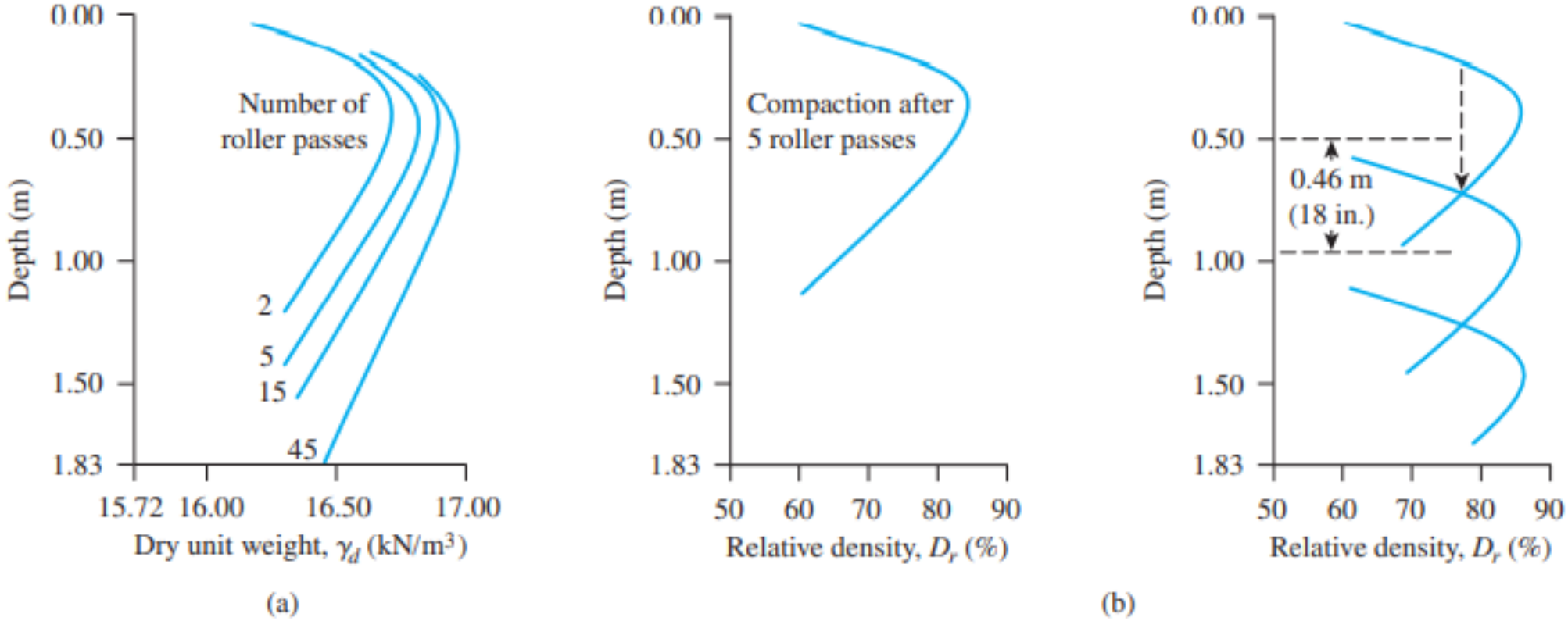


Figure 6.21 (a) Vibratory compaction of a sand—variation of dry unit weight with number of roller passes; thickness of lift = 2.44 m; (b) estimation of compaction lift thickness for minimum required relative density of 75% with five roller passes (*After D'Appolonia, Whitman, and D'Appolonia, 1969. With permission from ASCE.*)

Specifications for Field compaction

- field dry unit weight of 90% - 95% of the maximum dry unit weight determined in the laboratory by either the standard of modified Proctor test.

$$R(\%) = \frac{\gamma_{d(\text{field})}}{\gamma_{d(\text{max—lab})}} \times 100$$

- In the compaction of granular soils specifications are sometimes written in terms of the required relative density ***Dr*** **or compaction.**

$$D_r = \left[\frac{\gamma_{d(\text{field})} - \gamma_{d(\text{min})}}{\gamma_{d(\text{max})} - \gamma_{d(\text{min})}}\right]\left[\frac{\gamma_{d(\text{max})}}{\gamma_{d(\text{field})}}\right]$$

Specifications for Field compaction

$$R = \frac{R_0}{1 - D_r(1 - R_0)}$$

where $$R_0 = \frac{\gamma_{d(\min)}}{\gamma_{d(\max)}}$$

Based on the observation of 47 soil samples, Lee and Singh (1971) gave a correlation between R and Dr for granular soils:

$$R = 80 + 0.2D_r$$

Compaction of organic soil and waste materials

- The presence of organic materials in a soil reduces its strength
- High organic content are generally discarded as fill materials
- Soils with organic contents higher than about 10% are undesirable for compaction work
- standard procedures for determining the field unit weight of compaction include:

1. Sand cone method
2. Rubber balloon method
3. Nuclear method

Sand Cone Method(ASTM Designation D-1556)

- Very uniform dry Ottawa sand
- The weight of the jar, the cone, and the sand filling the jar is determined ($W1$)
- In the field, a small hole is excavated in the area where the soil has been compacted.
- dry weight of the soil $W3$

$$W_3 = \frac{W_2}{1 + \frac{w\,(\%)}{100}}$$

where w = moisture content

$W2$ = weight of moist soil excavated

- After excavation of the hole, the cone with the sand-filled jar attached to it is inverted and placed over the hole



Figure 6.23 Glass jar filled with Ottawa sand with sand cone attached (*Courtesy of Braja M. Das, Henderson, Nevada*)

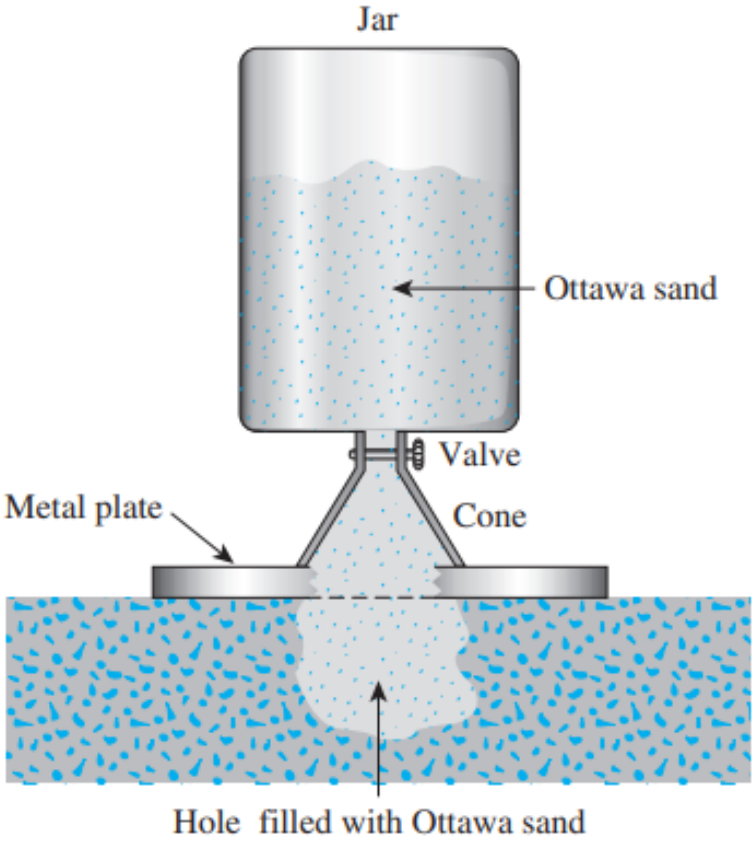


Figure 6.24 Field unit weight determined by sand cone method (*Most economical compaction condition*)

- $W5 = W1 - W4$
- where $W4$ = the weight of the jar, cone, and the remaining sand in the jar.
- $W5$ = weight of sand to fill the hole and cone.
- the volume of the hole excavated

$$V = \frac{W_5 - W_c}{\gamma_{d(\text{sand})}}$$

where Wc = weight of sand to fill the cone only

γd ($s$and) = dry unit weight of Ottawa sand used

$$\gamma_d = \frac{\text{Dry weight of the soil excavated from the hole}}{\text{Volume of the hole}} = \frac{W_3}{V}$$

- Nuclear Method
- The instrument measures the weight of wet soil per unit volume and also the weight of water present in a unit volume of soil



Figure 6.26 Nuclear density meter (*Courtesy of Braja M. Das, Henderson, Nevada*)

Summary and General Comments

- In this chapter, we discussed the following topics:

1. Laboratory compaction tests and related standards
2. Field compaction equipment and special field compaction techniques
3. Procedures used for determination of field unit weight compaction

Laboratory standard and modified Proctor compaction tests described in this chapter are essentially impact or dynamic compaction soil

However, In the laboratory, static compaction and kneading compaction can also be used.